

Data

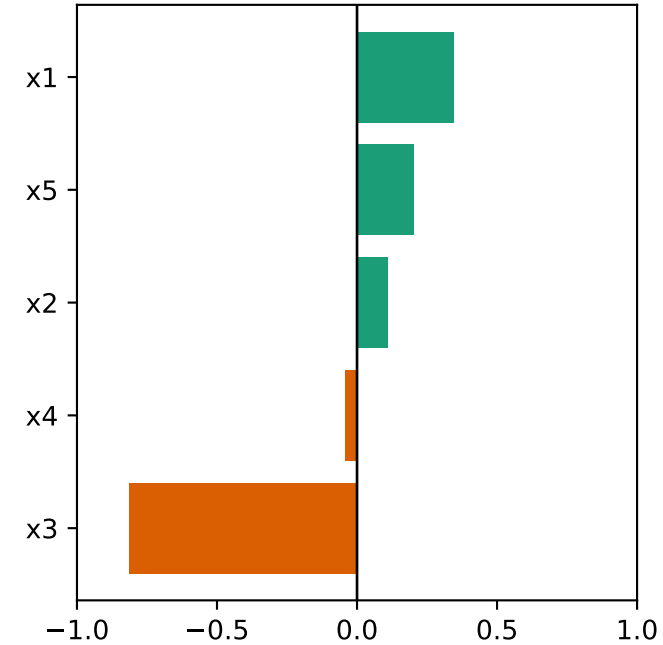
Sample Size & Structure

Training observations: 200
Test observations: 100
Predictors: x1, x2, x3, x4, x5
Response: y (continuous)

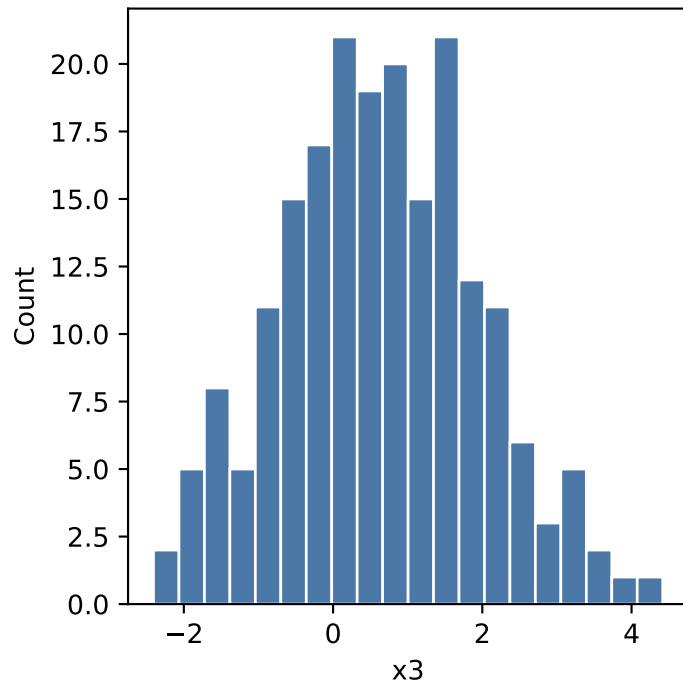
Summary Statistics

	x1	x2	x3	x4	x5	y
mean	0.28	-0.65	0.63	0.33	-0.20	0.74
std	1.08	1.00	1.31	1.05	0.79	2.25
min	-2.26	-3.78	-2.41	-2.53	-2.82	-4.62
max	2.63	2.18	4.42	3.06	2.07	6.41

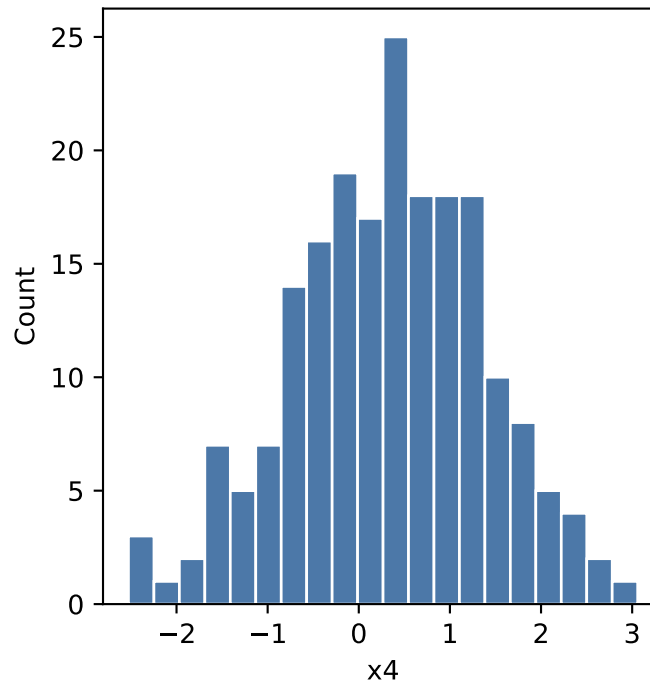
Correlation With y



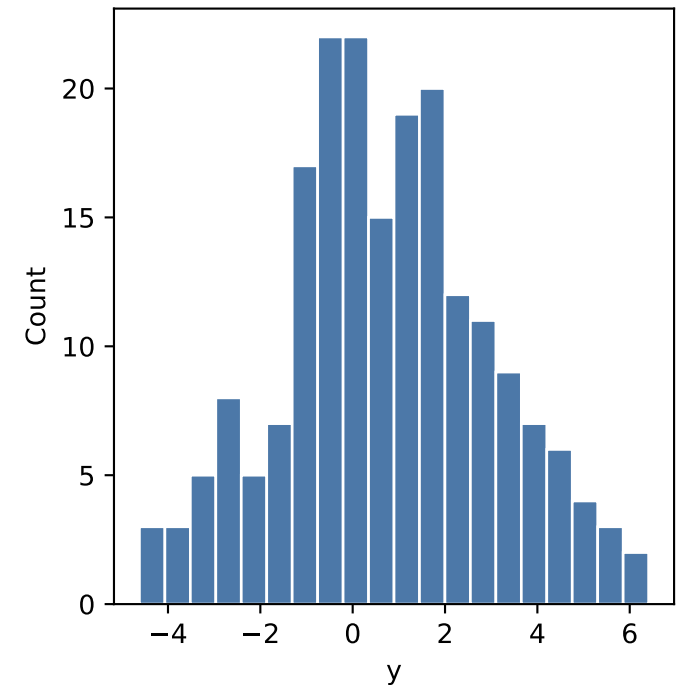
Histogram of x3



Histogram of x4



Histogram of y



Model, Method and Results

Model & Assumptions

Model:
y = beta_0 + sum(beta_j * x_j) + epsilon
use centered x3, x4, and x3*x4
epsilon ~ N(0, sigma^2)

- Assumptions:
- 1. linear mean structure
 - 2. independent observations
 - 3. constant error variance
 - 4. approximately normal residuals

Method

Method:
ordinary least squares
center x3 and x4 before interaction

Packages:
pandas
numpy
scipy.stats
matplotlib

Variable Selection

Full-model p-values:
x1: p=0.0000 (keep)
x2: p=0.3631 (drop)
x3: p=0.0000 (keep)
x4: p=0.7743 (drop)
x5: p=0.0000 (keep)
x3_c*x4_c: p=0.0120 (keep)

x4_c kept by hierarchy
Selected: x1, x3_c, x4_c, x5, x3c_x4c

Estimated Model

y_hat = 0.639 + (0.736)*x1 + (-1.391)*x3_c
+ (0.007)*x4_c + (0.513)*x5 + (0.118)*x3c_x4c

sigma_hat = 0.9526

Coefficients:

- intercept: 0.6390 (SE 0.0717)
- x1: 0.7362 (SE 0.0625)
- x3_c: -1.3912 (SE 0.0516)
- x4_c: 0.0072 (SE 0.0644)
- x5: 0.5130 (SE 0.0857)
- x3c_x4c: 0.1178 (SE 0.0464)

Coefficient Tests

name	coef	t	p
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intercept	0.639	8.91	0.0000
x1	0.736	11.78	0.0000
x3_c	-1.391	-26.96	0.0000
x4_c	0.007	0.11	0.9110
x5	0.513	5.98	0.0000
x3c_x4c	0.118	2.54	0.0120

Fit Summary

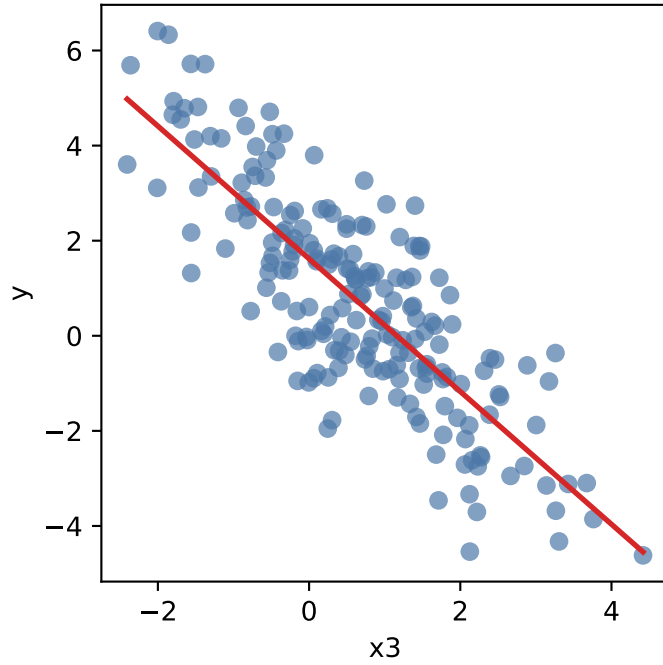
Model comparison:

spec	adjR2	AIC	BIC
additive	0.817	556.7	569.9
centered no x4	0.822	552.1	568.6
hierarchical*	0.821	554.1	573.9

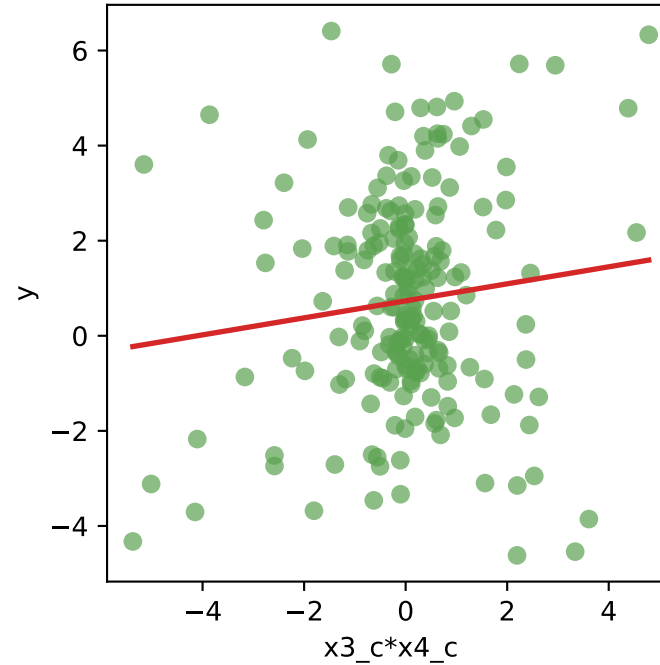
x4_c p-value: 0.9110
kept to preserve hierarchy
with centered x3_c*x4_c

Visualization

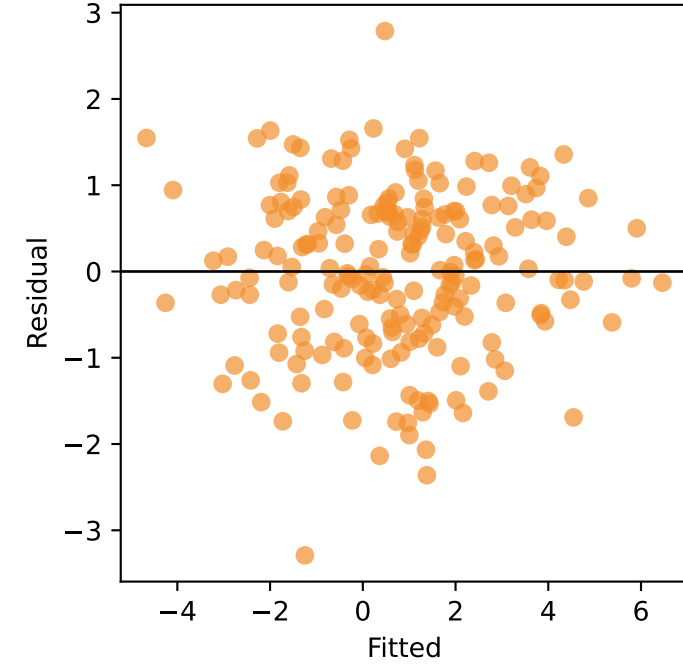
y vs x3



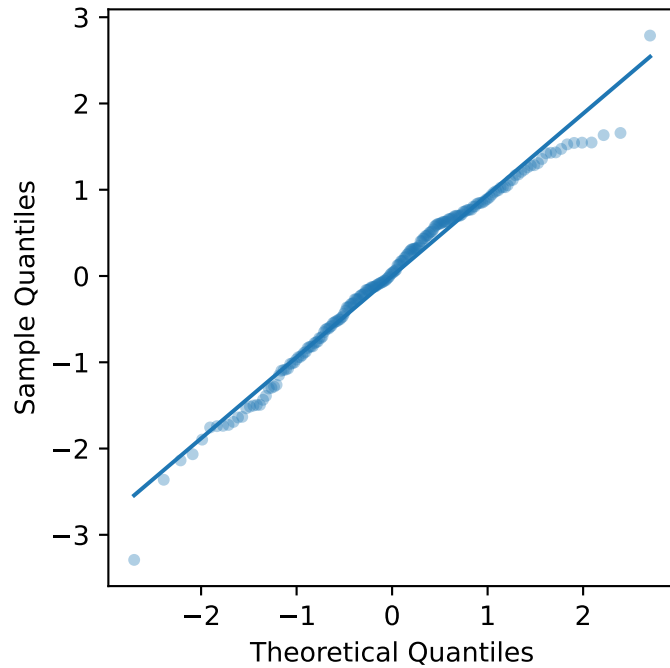
y vs x3_c*x4_c



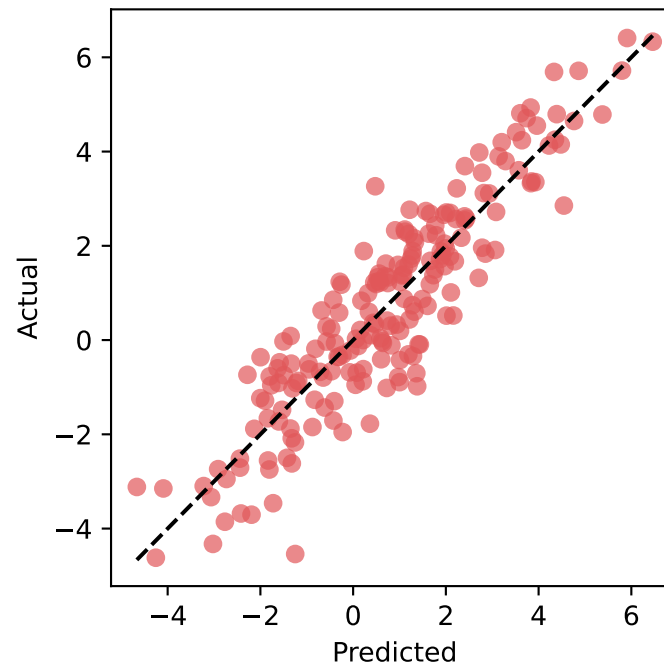
Residuals vs Fitted



QQ Plot



Predicted vs Actual



Coefficient Size

